

ArkLib’s KZG evaluation-binding theorem is axiom-clean **and** carries no information at any parameter: its t-SDH assumption quantifies over an unrestricted adversary type, which a `Classical.choice` trapdoor extractor inhabits with success probability exactly 1. We mechanize the refutation, the extraction-shaped repair that provably survives the exact attack, and generic-group numeric bounds — with the mechanization boundary stated scrupulously.

Draft — internal working paper, not filed, not a security advisory: a **formalization-soundness** issue in a public, in-development library, not a vulnerability in any deployed system. KZG, t-SDH, and the reduction are sound as normally stated — the issue is a Lean quantifier; nothing here is embargoed or exploitable.

I · The finding: axiom-clean, and empty

The mechanism

KZG evaluation binding is proved in ArkLib by a correct, constructive reduction to the t-SDH assumption — whose adversary is a **plain, unrestricted function type**:

```
abbrev tSdhAdversary (D : ℕ) :=
  Vector G₁ (D+1) × Vector G₂ 2 → StateT unifSpec.QueryCache ProbComp (Option (ZMod p × G₁))

def tSdhAssumption (D : ℕ) (error : ℝ≥0) : Prop :=
  ∀ (adversary : tSdhAdversary D), tSdhExperiment D adversary ≤ error
```

ProbComp is a free monad over oracle queries: only `query` nodes cost anything; pure computation is free and no resource bound is imposed anywhere. The SRS includes the verifier leg (g_2, g_2^τ) , which determines τ whenever $g_2 \neq 1$, and ArkLib’s own `exists_zmod_power_of_generator` (`Algebra.lean:105`) makes the discrete log `Classical.choice`-definable. The adversary reads g_2^τ , recovers τ , and returns the t-SDH solution $(c = 0, g_1^{1/\tau})$ — zero oracle queries, success probability exactly 1.

```
noncomputable def tauExtractingAdversary (hg₂ : g₂ ≠ 1) (D : ℕ) : tSdhAdversary D :=
  fun srs => pure (some (0, g₁ ^ (1 / dlogOf hg₂ srs.2[1])).val))

theorem tSdhExperiment_tauExtractingAdversary : tSdhExperiment D (...) = 1
theorem not_tSdhAssumption (herr : error < 1) : ¬ tSdhAssumption D error
theorem tSdhAssumption_trivial_of_one_le (1 ≤ error) : tSdhAssumption D error
```

No content at any parameter

Below 1 the assumption is refuted; at ≥ 1 its conclusion is the triviality “a probability is ≤ 1 .” `binding` consumes the assumption in the last step of its `calc`, and its hypothesis `hpair : pairing g₁ g₂ ≠ 0 forces g₂ ≠ 1` (a bilinear map kills the identity) — exactly what the extractor needs. So `binding`’s hypotheses are jointly unsatisfiable for any error below 1 (`binding_hypotheses_unsatisfiable`), and its conclusion is free at ≥ 1 . The sibling ARSDH assumption falls identically, taking `function_binding` with it. A canary — the giving-up adversary scores exactly 0 — confirms the experiment discriminates. The reduction itself is fully constructive and algebraically correct; the vacuity lives entirely in the assumption’s quantifier.

The pattern, not a typo

Reducing to a **different** base assumption does not escape. The natural q-DLOG assumption, stated in ArkLib’s own unrestricted-adversary idiom, is equally false below 1 (`not_qDlogAssumption`, same extraction, own canary). And ArkLib’s algebraic-group-model scaffolding confirms the disease from the other side: `AGM/Basic.lean` is a stub (its adversary’s `run` is literally `sorry`; zero theorems; orphaned) and is **unsound as written** — the adversary is a `ReaderT` over the concrete group table, so its outputs can still depend on discrete logs. Its author flags the open problem verbatim: “TODO: need to be sure this definition is correct” “How to make the adversary truly independent of the group description? It could have had G hardwired.” Any concrete-group assumption of the shape $\forall$ unrestricted adversary, $\Pr[\text{win}] \leq \varepsilon < 1$ is `Classical.choice`-false in this idiom.

The methodological point: axiom checks are blind to vacuity

`binding`, the refutation, and the winning adversary all print the **same** clean axiom closure:

```
#print axioms not_tSdhAssumption -- [propext, Classical.choice, Quot.sound]
```

No `sorryAx`, nothing any `#print axioms / #assert_axioms` gate would flag. **Axiom-clean and vacuous coexist**. The blindness is structural: an axiom check reports a proof term’s closure; it never asks whether the theorem’s **hypotheses** are jointly satisfiable, and any `def FooHard : Prop` used as a hypothesis is an assumption no axiom check inspects. Query-based bounds (`IsQueryBoundP`) do not help either — the winning adversary makes zero queries. The only reliable test is adversarial: try to **inhabit the assumption’s negation** — prove the floor false at its deployed parameters. We found the identical pattern in our own hardness floors first, in several places, before ever looking at ArkLib; we present this as a field lesson, not a dunk.

II · The repair, the number, and the honest status

The mergeable de-vacuation (mechanized)

ArkLib’s reduction is already constructive, and the assumption is consumed at exactly one `calc` step — so split there. The unconditional prefix becomes the primary theorem; the original `binding` becomes a one-line corollary.

```
theorem binding_reduces_to_tSdh ... (adversary : KzgBindingAdversary ...) :
  bindingExperiment ... adversary
    ≤ tSdhExperiment g₁ g₂ n (bindingReduction ... adversary)
```

No assumption `Prop`: both sides are concrete probabilities, true at every parameter — nothing for `Classical.choice` to inhabit. The diff is +41/-14 in one file; the whole tree builds (2994 jobs); axiom-clean. And it **provably survives the exact attack** (`repair_survives_attack`): in one `sorry`-free closure, (1) the trapdoor adversary still refutes `tSdhAssumption` below 1, and (2) the repaired bound holds regardless. The cure removes the unsatisfiable premise while keeping every step of the reduction. It does not by itself supply a number — it isolates the one obligation a sound assumption class must discharge.

The number: the generic bilinear group bound

Group elements become opaque handles carrying **ordinary** polynomials in $\mathbb{Z}_p[X]$ — **not Laurent**: group inversion negates the exponent, it never introduces X^{-1} . A winning $1/(X + c)$ is therefore unrepresentable, and a “win” forces the nonzero, degree- $\leq D + 1$ polynomial $F_\ell \cdot (X + c) - 1$ to vanish at the random τ . Simulation (identical-until-bad) plus Schwartz–Zippel yield Boneh–Boyen’s Theorem 12, verified line by line against the source:

$$\varepsilon \leq (q_G + D + 3)^2 \frac{D + 1}{p - 1} = O((q_G + D)^2 \cdot D/p)$$

Not a clean q^2/p : the bound is cubic in the SRS degree D , and at production parameters the D^3/p term is the one to watch (Corollary 13’s $q < O(p^{1/3})$ side condition). Naive AGM is not a shortcut: an adversary that **also** returns a valid representation is still `Classical.choice`-inhabitable — validity is not independence — so the AGM route relocates the same content onto q-DLog’s generic hardness.

Sorry-free in Lean today (all axioms [propext, Classical.choice, Quot.sound])

- **Vacuity refutations** — t-SDH, ARSDH, q-DLOG, each with a discriminating canary; imports genuine ArkLib at d72f8392, redefines nothing.
- **The repair and its survival** — `binding_reduces_to_tSdh` and `repair_survives_attack`.
- **Static GGM bound** — `ggm_tSdh_sound`: $\varepsilon \leq (D + 1)/(p - 1)$ over the **entire** committed-generic adversary type (offset + degree- $\leq D$ polynomial, no group input — the trapdoor extractor is untypable here). Tight; $\approx 2^{-234}$ at $p \approx 2^{254}$, $D \approx 2^{20}$. As far as our census found, the first generic-group security theorem in Lean.
- **Adaptive bound, explicit-equality-oracle (Maurer) model** — `adaptive_ggm_sound`: $\varepsilon \leq (\text{fuel} \cdot \Delta + (D + 1))/(p - 1)$ under adaptive group-op / pairing / equality queries; identical-until-bad **proven by induction** (`runAux_congr_of_agree`), not assumed; `fuel = 0` recovers the static bound. Linear in equality queries — a tighter model than Shoup’s, where comparison is free.
- **Quadratic Shoup (random-encoding) bound** — `rand_encoding_bound`: $\varepsilon \leq (\binom{n}{2} \cdot 2D + (D + 1))/(p - 1)$, $n = \text{fuel} + D + 4$; the all-pairs bad event and the table size are theorems, not assumptions.
- **Degree invariant, structural** — $2D$ under the two-sorted pairing discipline; the naive flat-table $2D$ claim **refuted** (X^4 at $D = 1$). Proved for a peer model.
- **ArkLib condition transport** — `groupWinSet_eq_realWinSet`: the field-level win predicate is ArkLib’s real `tSdhCondition`, by prime-order injectivity.

The named frontier (open — stated exactly, not gating the above)

- **Degree hypotheses undischarged**. The adaptive and random-encoding theorems still consume `hdeg_*` as hypotheses. The structural $2D$ proof lives in a peer model (`PairedOp/buildPaired`) that the oracle `runAux` does not import — and it is a **genuine restriction** the oracle must adopt, not bookkeeping: the flat oracle provably violates $2D$. Until wired in, those theorems are not hypothesis-free.
- **ProbComp plumbing**. Threading the condition-level transport into the literal `tSdhExperiment` inequality (the `Strat → tSdhAdversary` embedding, the sampler’s $\Pr = \text{card}/(p - 1)$ semantics), and re-typing `bindingReduction`’s constructed adversary as a straight-line `Strat` — so `binding` itself inherits the bound.
- **The interlock is architectural, not mechanized**. The chain *random-encoding bound* $\leftarrow$ *degree* $\leq 2D \leftarrow$ *pairing discipline* is designed, with each link proven in its own file; the bridge lemmas between them remain to be written.